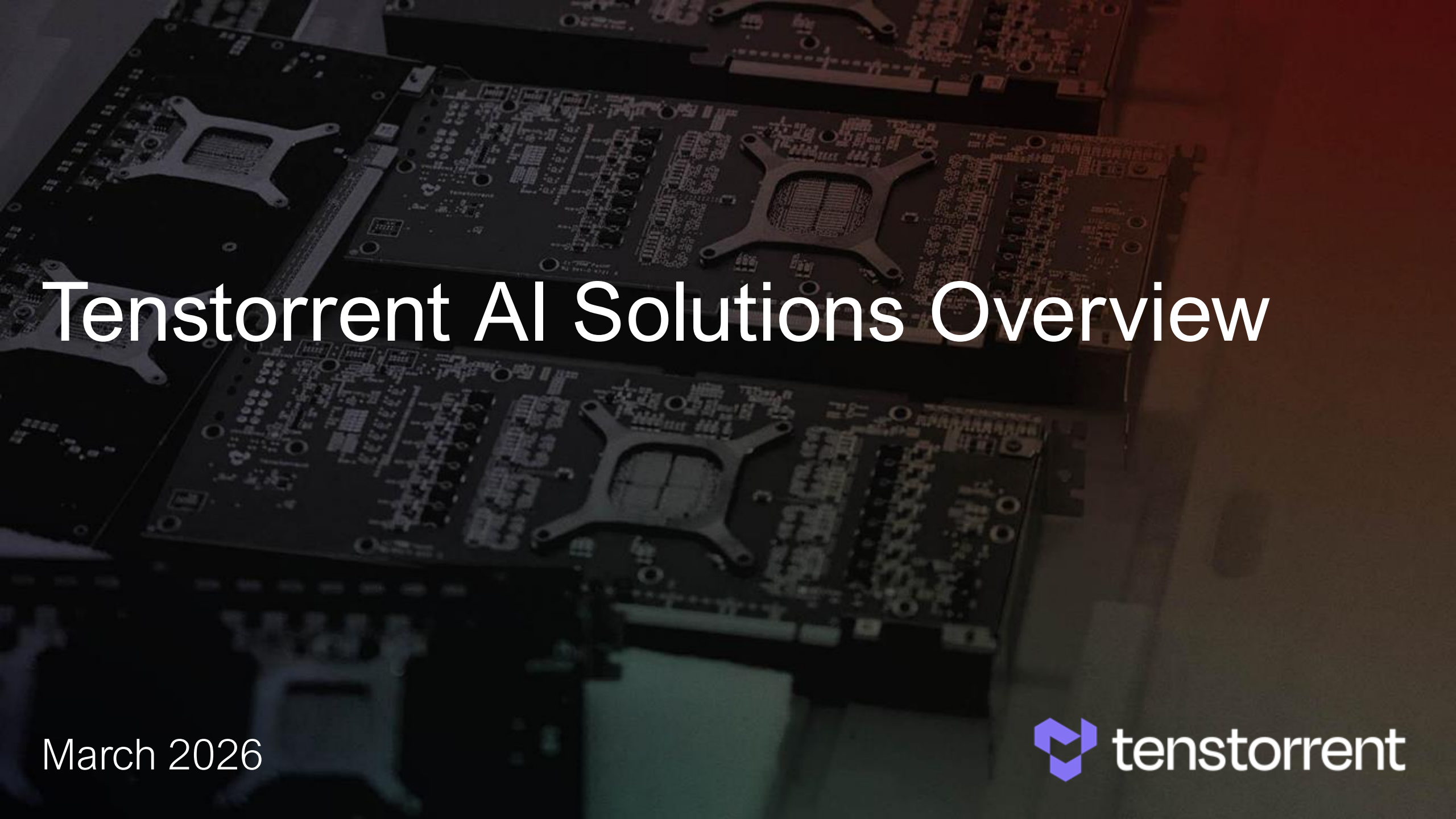




Tenstorrent AI Solutions Overview

March 2026



Agenda

Tenstorrent Introduction

Hardware

Software

AI Models

Scale-Out

Tenstorrent at a Glance



Founded
March 2016



Capital Raised
\$1B+

Investors



EPIQ



Target Markets



Data Center



Automotive /
Robotics



Edge AI / IoT

\$400B

AI / Compute
TAM

Key Performance Indicators

~1,000

Employees

700+

R&D
Professionals

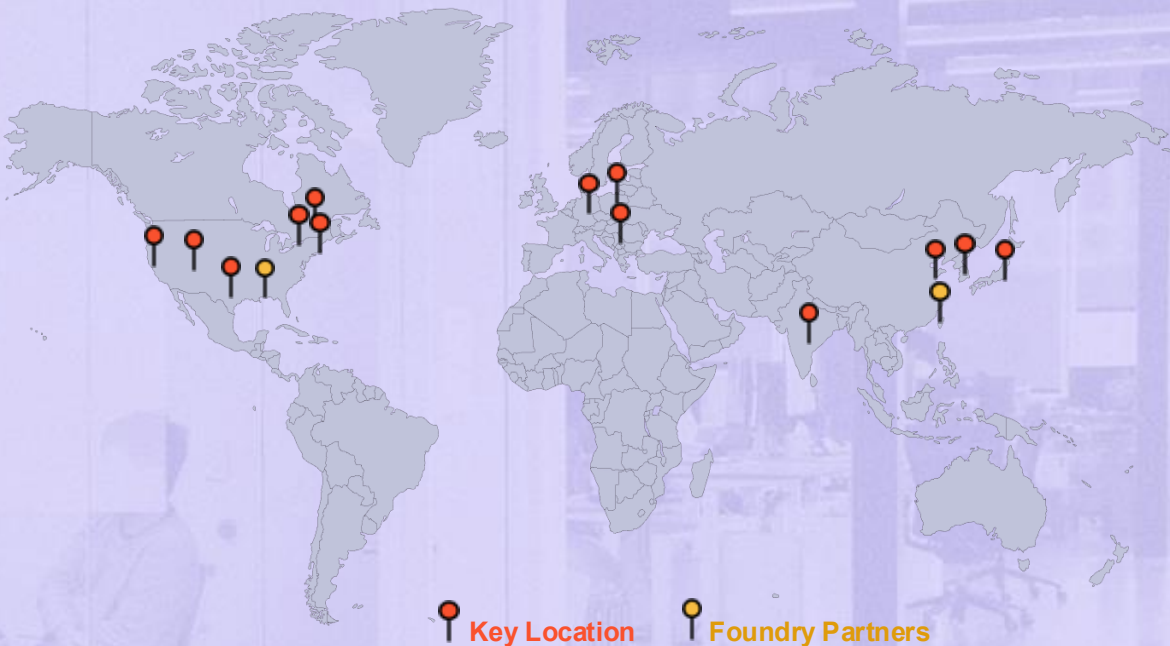
900+

Developers on
TT platform

~\$250M+

2026E
Revenue

Global Footprint



Business Model

Tenstorrent monetizes its AI and RISC-V Technology in 3 ways:

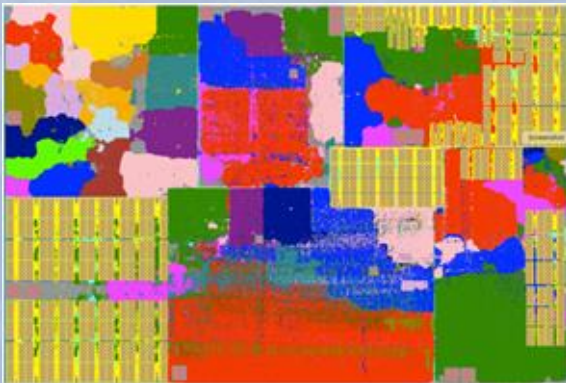
1. Licenses AI and RISC-V IP to customers building silicon
2. Sells AI Chiplets, RISC-V Chiplets, & AI Chips
3. Sells AI boards, servers and systems

TT Mission: Make AI & HPC Compute Open & Accessible

Open, fast & cheap AI is our North Star & guides everything we do, from sys design choices to how we do business

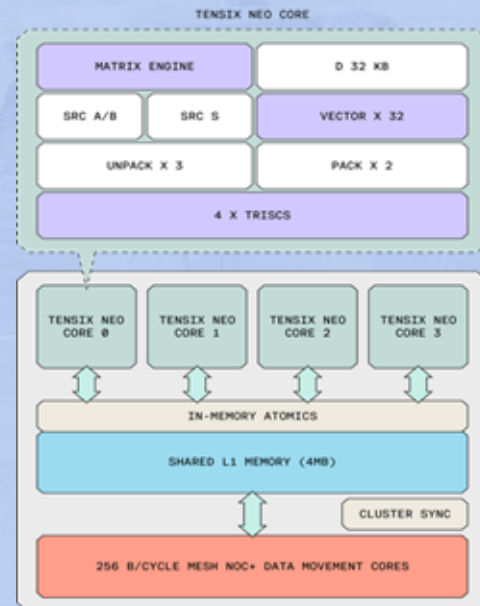
RISC-V based Open Architecture

- **NO monopoly ISA fees!**
- Fastest growing compute ecosystem



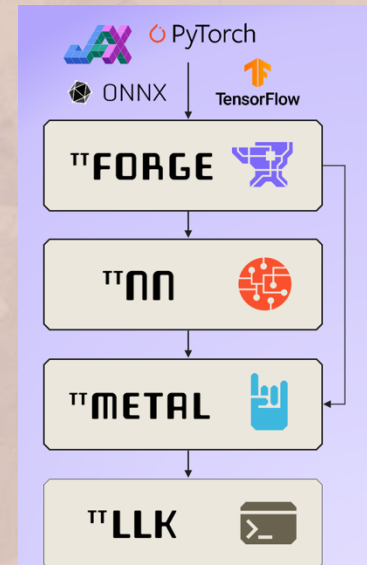
Open Tensix AI Processor

- **NO vendor lock-in!**
- Fair & friendly business model



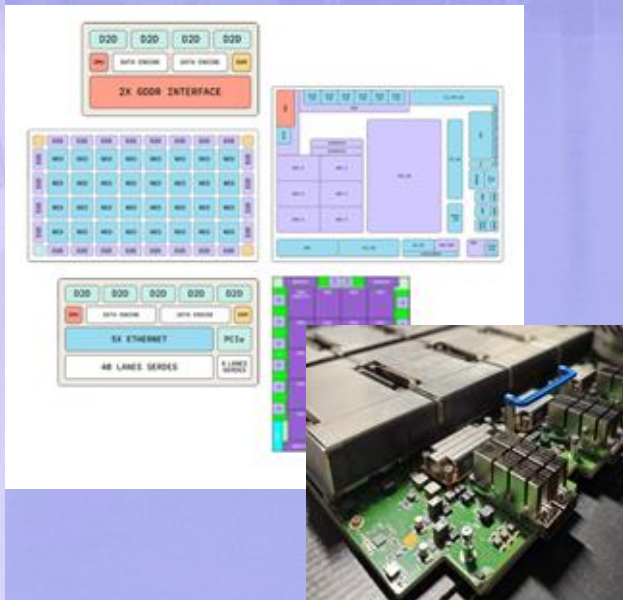
Open-Source AI Software

- **NO proprietary stacks!**
- Full transparency & customization flexibility



High Performance AI with GDDR & Ethernet

- **NO HBM or proprietary interconnect!**
- We save your \$\$\$!



Flexible Business Model

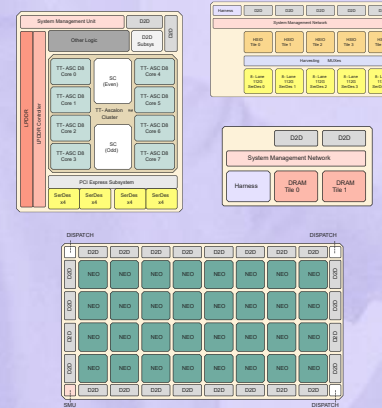
We meet our customers where they are, whether at IP, silicon, system or SW/service level

IP (TT-Ascalon™ / Tensix NEO)



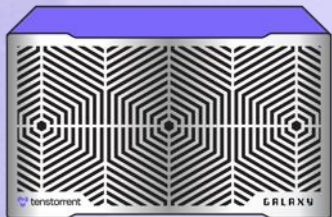
- Scales from mW to MW for efficiency and performance
- IP available for licensing
- Industry-leading performance
- Modular design available in varied configurations

Chips & Chiplets



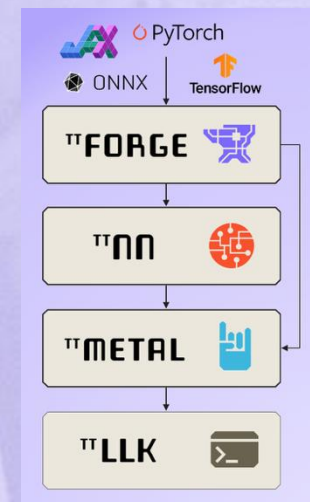
- Portfolio of products powered by scalable Tensix AI cores
- Inference and training, CNN and NLP, recommendation engines, all on the same silicon
- Hardware available for purchase as well as IP available for licensing
- Multi-component modular chiplets

Servers (Tenstorrent Galaxy™)



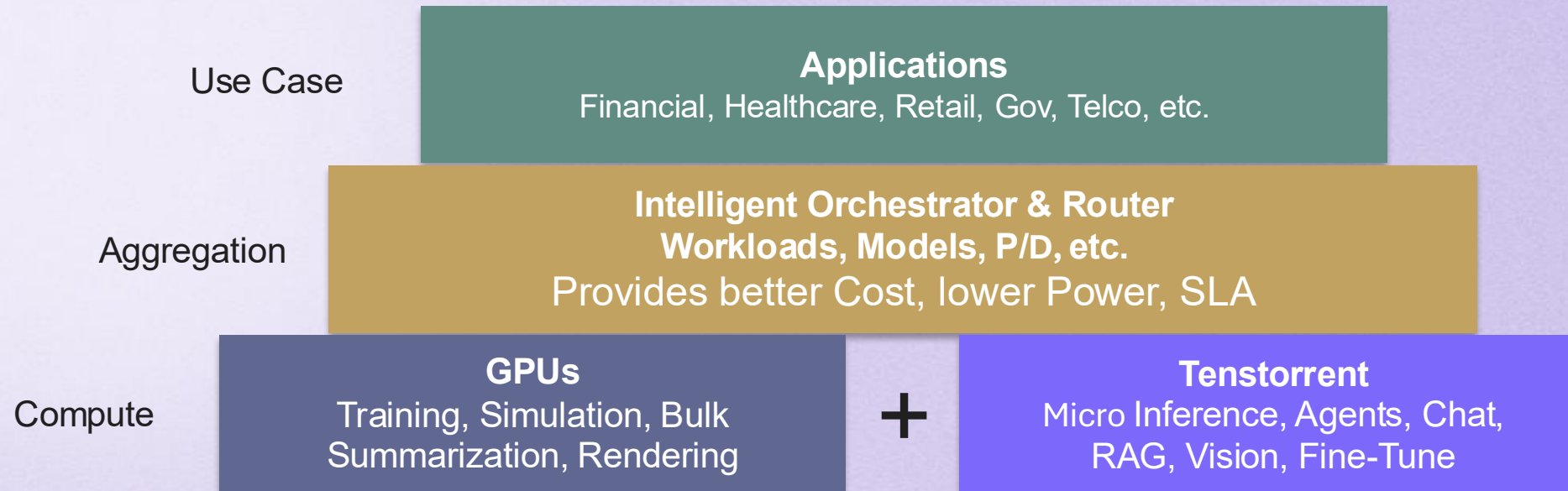
- Galaxy Server – 32 high performance ASICs in a custom chassis
- Easily combine servers into a Galaxy Rack with high bandwidth chip-to-chip connectivity

Software



- Fully open source software stack with multiple entry points
- TT-Forge™ - MLIR-based compiler (Q2 2025)
- TT-NN™ – Library of optimized operators
- TT-Metalium™ - Bare metal, low-level programming model

Heterogeneous Cloud Deployment



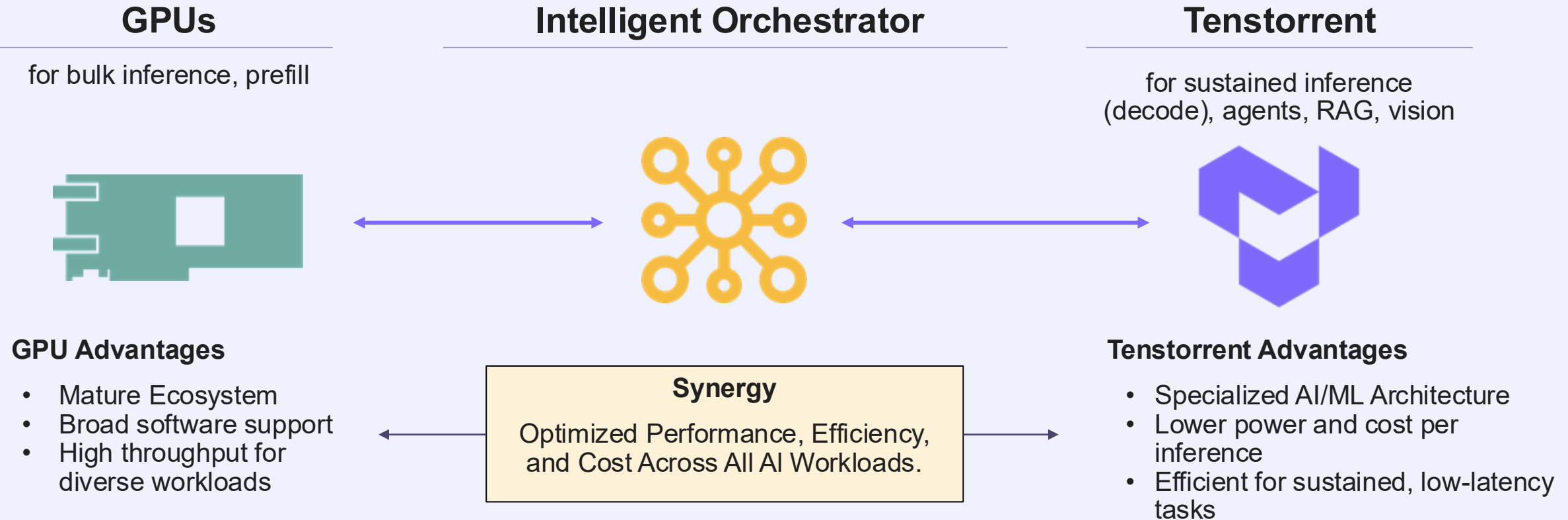
Tenstorrent HW can be implemented alongside GPUs – this is a huge scaling opportunity.

- Avoids long-term vendor lock-in and GPU oversizing risk
- Intelligent orchestrator routes workloads in real time
- GPUs for training and long-context prefill
- Tenstorrent for sustained inference, agents, RAG, vision
- Unified APIs, billing, and observability
- Linear TCO scaling as demand grows
- GPU count reduced 30–60% for inference-heavy clouds
- Higher margins for neo-clouds & sovereign operators

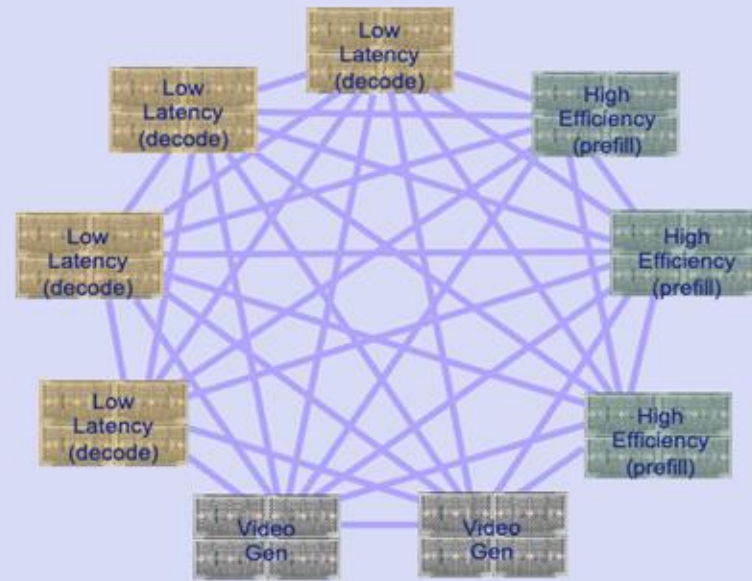
Heterogeneous Cloud Deployment - Compute Layer

Workload, Model and P/D Disaggregation

- Optimized resource utilization by separating prefill from decode
- Scalable performance through model parallelization across devices



Galaxy System: Building Blocks to Scale

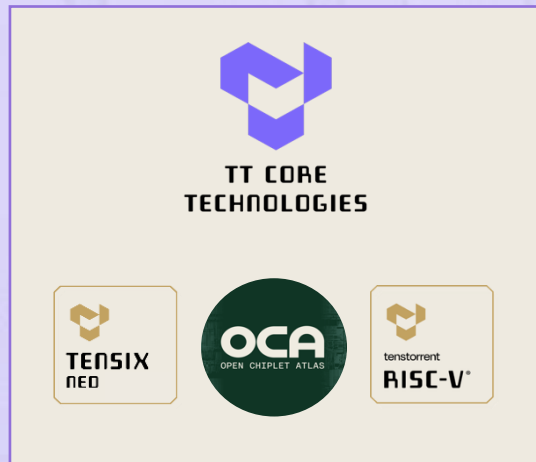


Highly Flexible Configuration Enables:

- Batch workloads to leverage high-throughput
- Realtime applications leveraging low-latency
- Realtime Video Generation

Multiple, connected 4-Galaxy nodes can uniquely deliver range of performance models in an integrated system without additional overhead

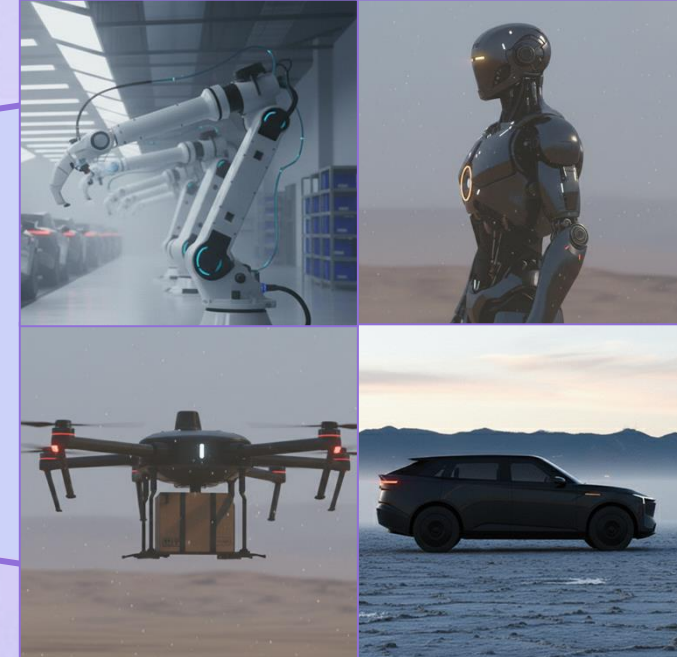
Scalable Architecture, from Servers to Robots



**Open Hardware
Architecture**

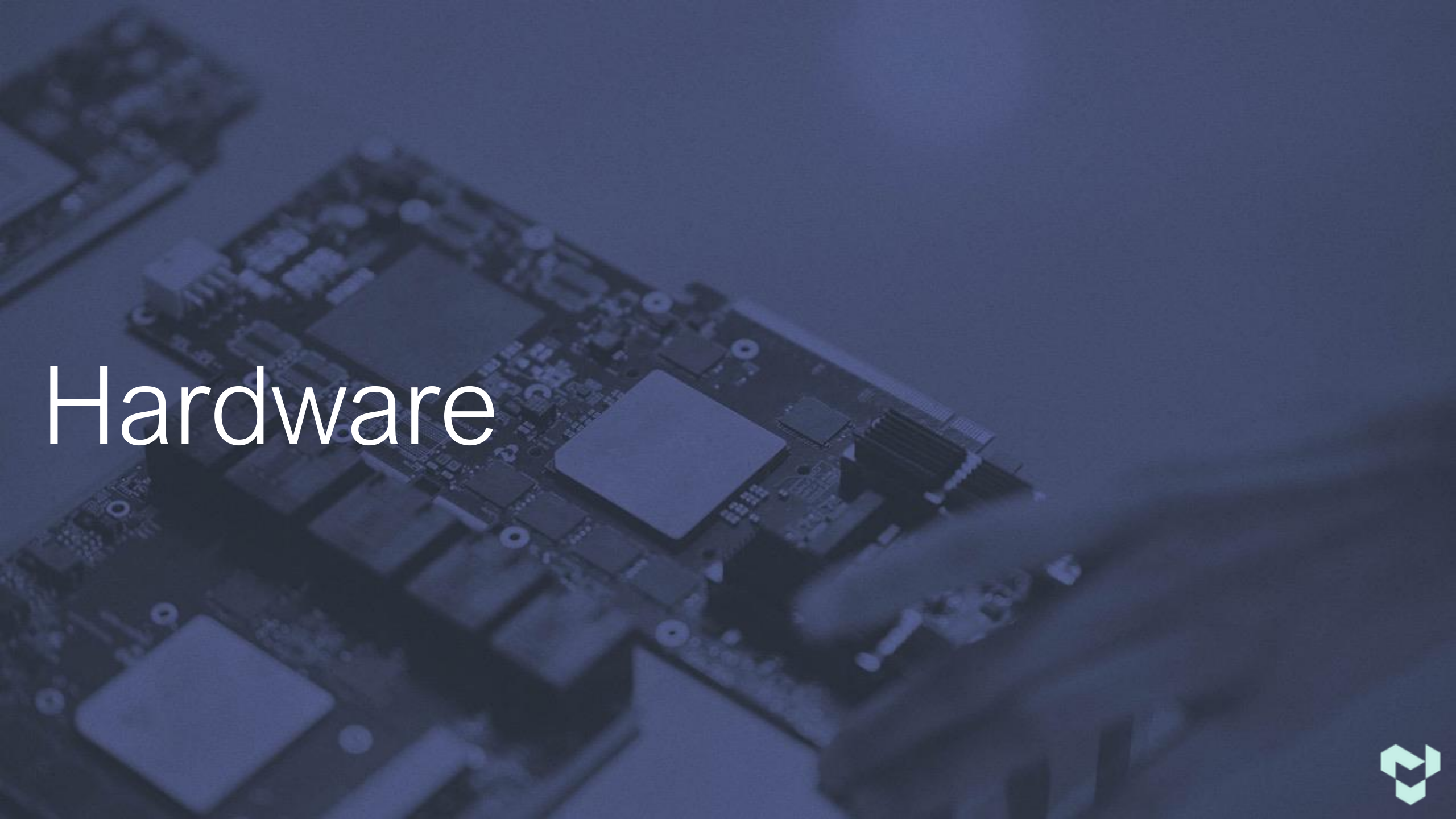


**High
performance
servers**



**Robotics &
Autonomous Systems**

Shared Open-Source Software and Tools | Edge-to-Cloud Integration



Hardware



Tenstorrent Hardware Portfolio



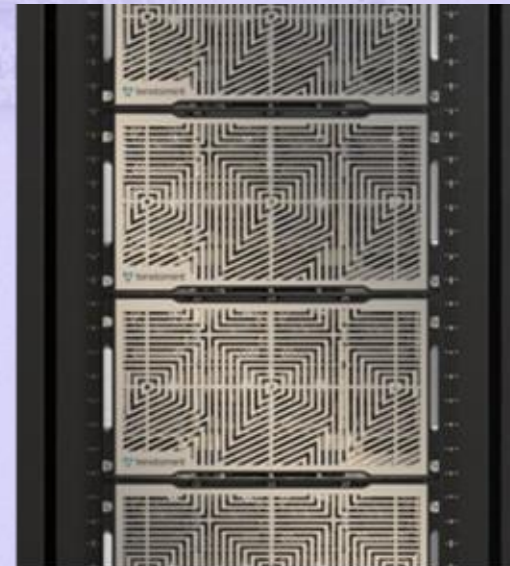
Cards



TT-QuietBox™



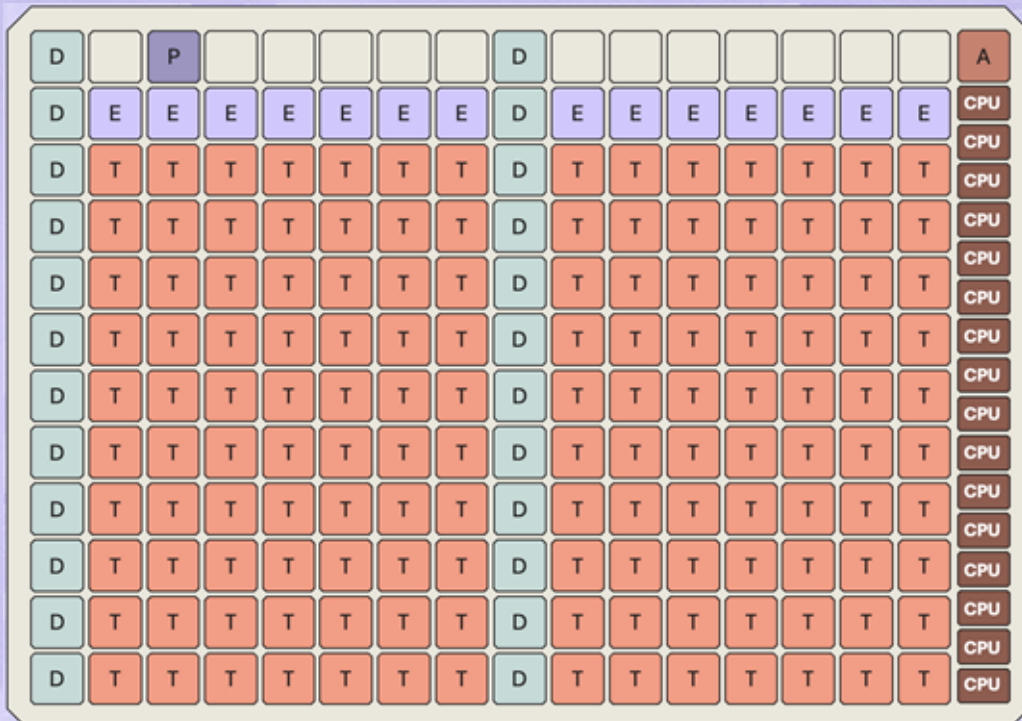
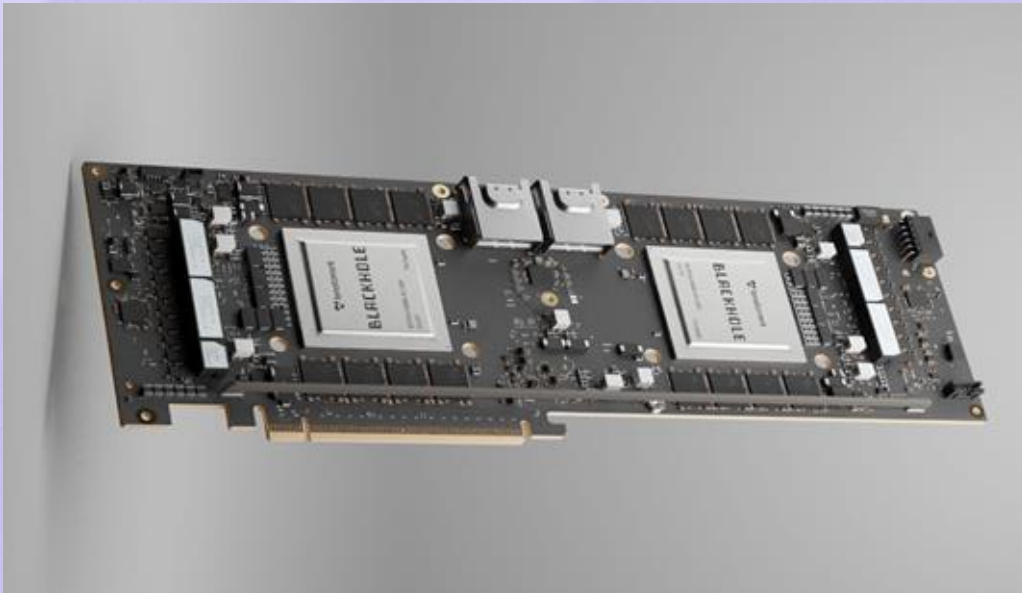
TT-LoudBox™



Tenstorrent Galaxy™



Blackhole™

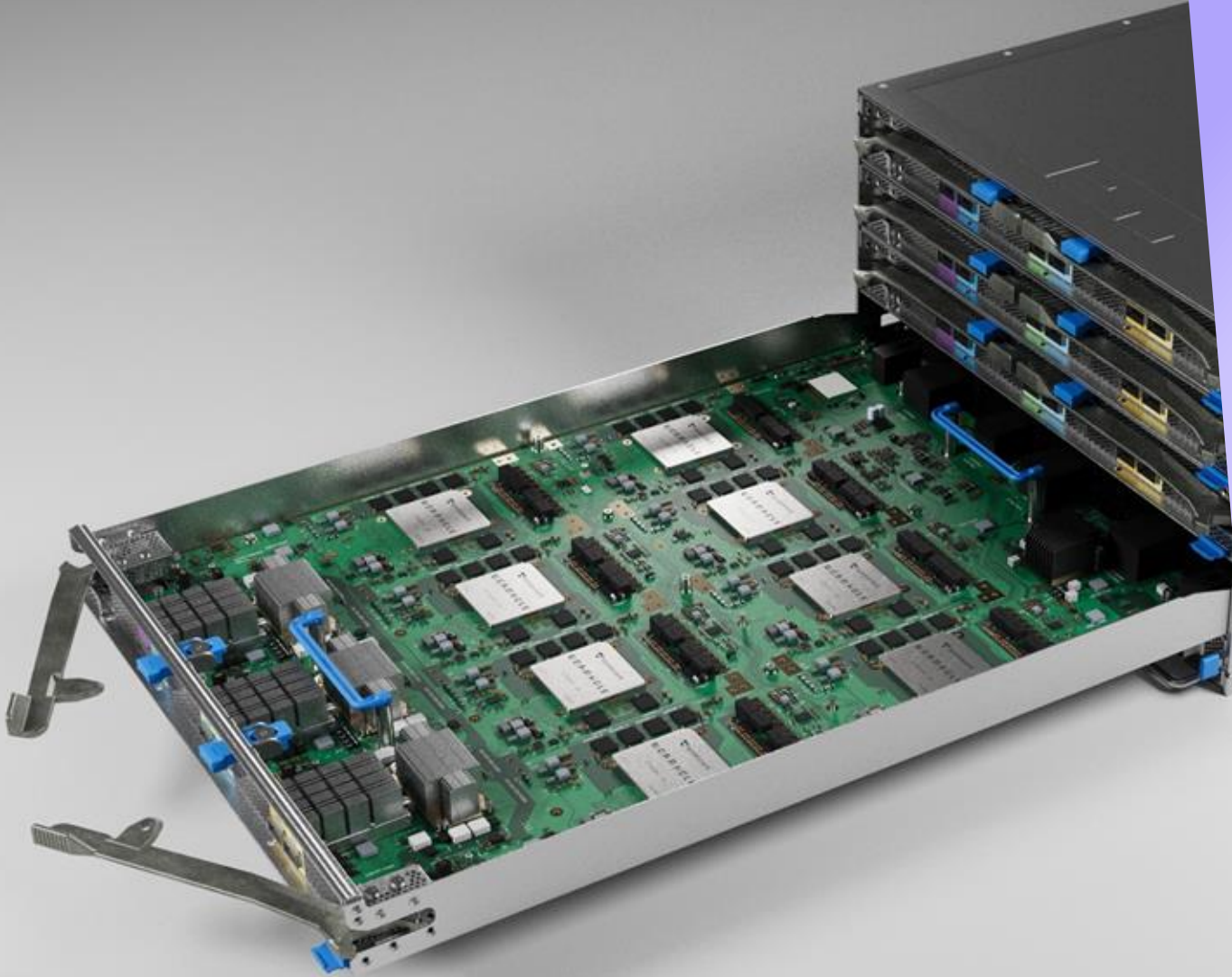


- CPU** RiscV CPUs
- T** Tensor Cores
- D** DRAM Cores
- E** ETH Cores
- P** PCIe Core
- A** ARC Core

- 120 Tensix AI Processors
- 664 TFLOPs (8-bit)
- 32GB GDDR6 @ 512 GB/s
- 180MB SRAM
- 16 Risc-V Processors
- Dual 512-bit 2D Torus NOC
- PCIe Gen5 x 16
- Up to 10 x 400 Gbps Interconnect for Scale-out

Blackhole Galaxy

- 32 Blackhole chips
- GDDR6: 1 TB @ 16 TB/s
- 56 x 800 Gbps Ethernet
- Unified scale-up & scale-out
- Virtualize multiple Galaxies into Exaflop-scale systems



Tenstorrent Edge

Differentiating Use Cases

Use Case Type	Compute-Bound Workloads Long Context, Large Matmul		Memory-Bound Workloads Fast & Deterministic Execution	
	Video Gen Models	LLM Prefill	Blitz Decode	High- Frequency
Very Large SRAM			✓	✓
Near-Deterministic Runtime			✓	✓
Persistent Kernels			✓	✓
High FLOPs Compute	✓	✓		
Low Cost DRAM Memory	✓	✓	✓	✓
Ultra-High BW & Flex Scale Out	✓	✓	✓	✓
Implications	Cost-effective scaling, right-sized to the model knee & not rack size		FPGA-like, fast & cost-efficient execution with massive SRAM	

- **Tenstorrent excels on compute-bound workloads with long contexts** such as Video Gen & LLM Prefill
- Tenstorrent is also **highly differentiated on memory-bound workloads** leveraging massive SRAM (6x of GPUs) such as LLM decode (blitz) & HFT
- **Ultra-high BW & flexible scale-out** (10x better) drive both cost-efficient & fast execution across all key use-cases

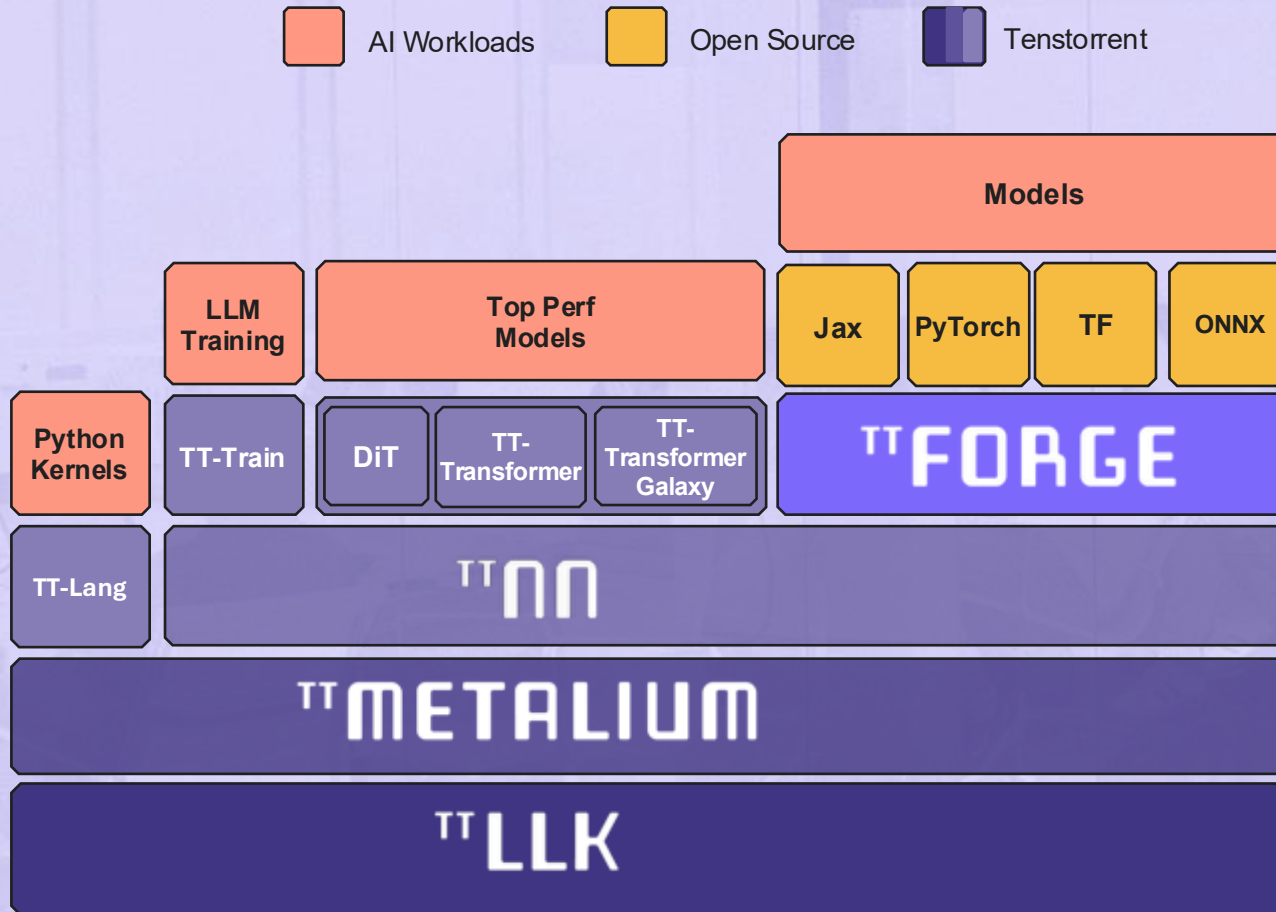


Software

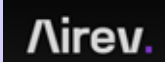


Building Block Approach

- Multiple entry points, from pytorch down to low-level kernel
- Fast growing ecosystem of partners & developers



Third-Party Developers

	Model Development
	AI Kernel Generation
	DX & Model Bring-up
	Model Development
	Model Development
	Triton compiler
	In-Game AI
	Cuda PyTorch Cross Compiler
	AI Agents
	Datacenter AI Visualization
	AI Training
	ML compiler
	ML framework

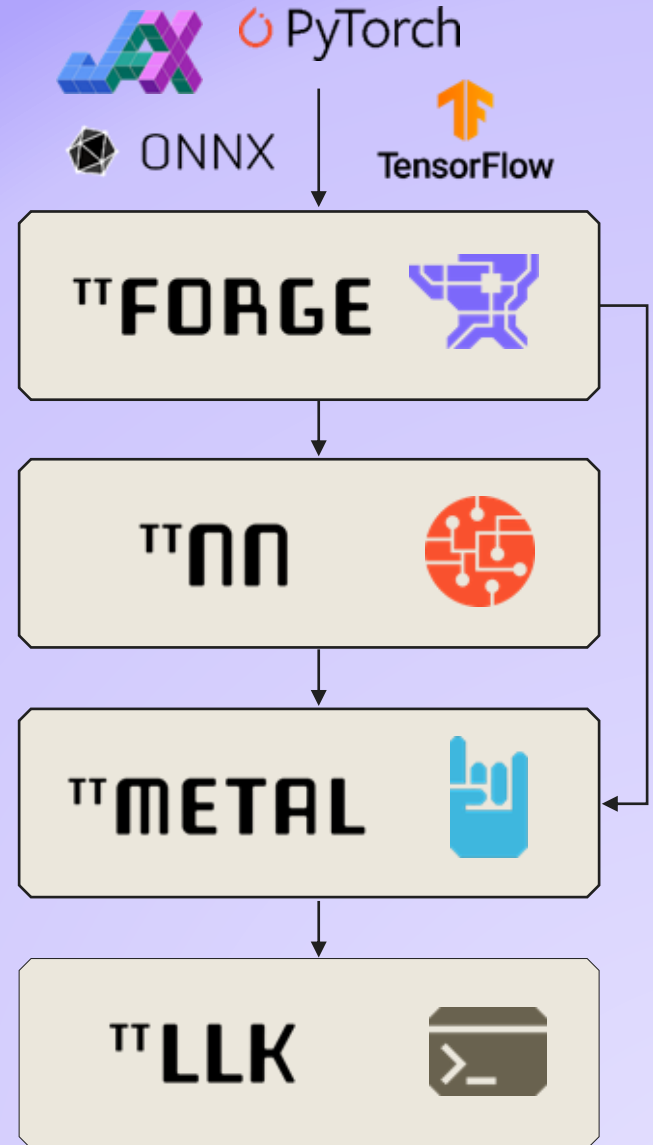
Open-Source Software Stack

TT-Forge is an MLIR based compiler that runs hundreds of models

TT-NN is a library of high-performance operators and models

TT-MetaliumTM is a kernel programming model

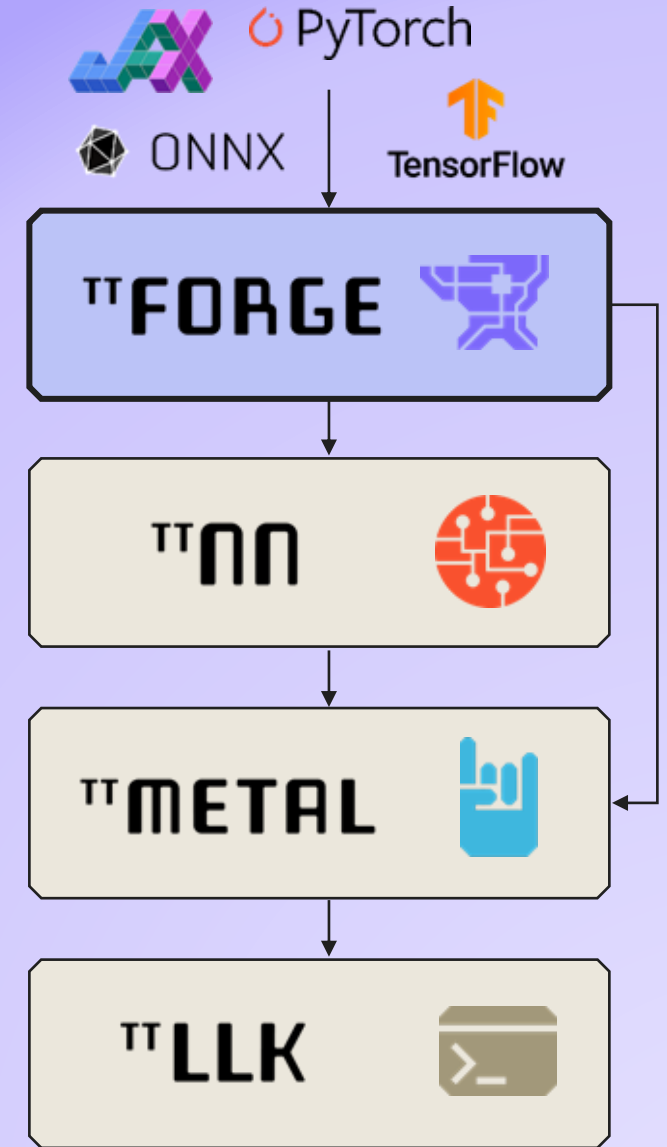
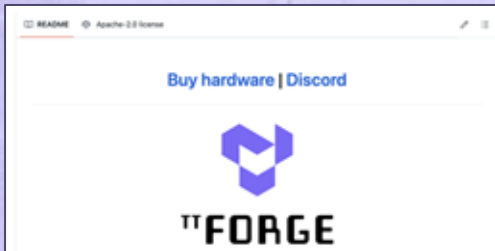
TT-LLK is a library of HW intrinsics



TT-Forge

- MLIR-based compiler integrated into various frameworks
- AI/ML models from domain-specific compilers to custom kernel generation

```
64 def run_resnet_pytorch(variant="microsoft/resnet-50", batch_size=1):
65
66     # Load ResNet feature extractor and model checkpoint from HuggingFace
67     model_ckpt = variant
68     feature_extractor = AutoFeatureExtractor.from_pretrained(model_ckpt)
69     model = ResNetForImageClassification.from_pretrained(model_ckpt)
70
71     # Load data sample
72     url = "https://images.rawpixel.com/image_1300/cHJpdmcF0ZS9sci9pbWFnZXMvd2Vic2l0ZS8yMDIyLw=="
73     image = Image.open(requests.get(url, stream=True).raw)
74     label = ["tiger"] * batch_size
75
76     # Data preprocessing
77     inputs = feature_extractor(image, return_tensors="pt")
78     pixel_values = [inputs["pixel_values"]] * batch_size
79     batch_input = torch.cat(pixel_values, dim=0)
80
81     # Run inference on Tenstorrent device
82     framework_model = ResNetForImageClassification.from_pretrained("microsoft/resnet-50")
83
84     # Compile the model using Forge
85     compiled_model = forge.compile(framework_model, batch_input)
86
87     output = compiled_model(batch_input)
88     # Data postprocessing
89     predicted_value = output[0].argmax(-1)
90     predicted_label = [model.config.id2label[pred.item()] for pred in predicted_value]
91
92     for sample in range(batch_size):
93         print(f"True Label: {label[sample]} | Predicted Label: {predicted_label[sample]}")
94
95 if __name__ == "__main__":
96     run_resnet_pytorch()
97
```



TT-NN

- TT-NN is a Python and C++ neural network operator library
- Library of optimized operators with PyTorch-like API

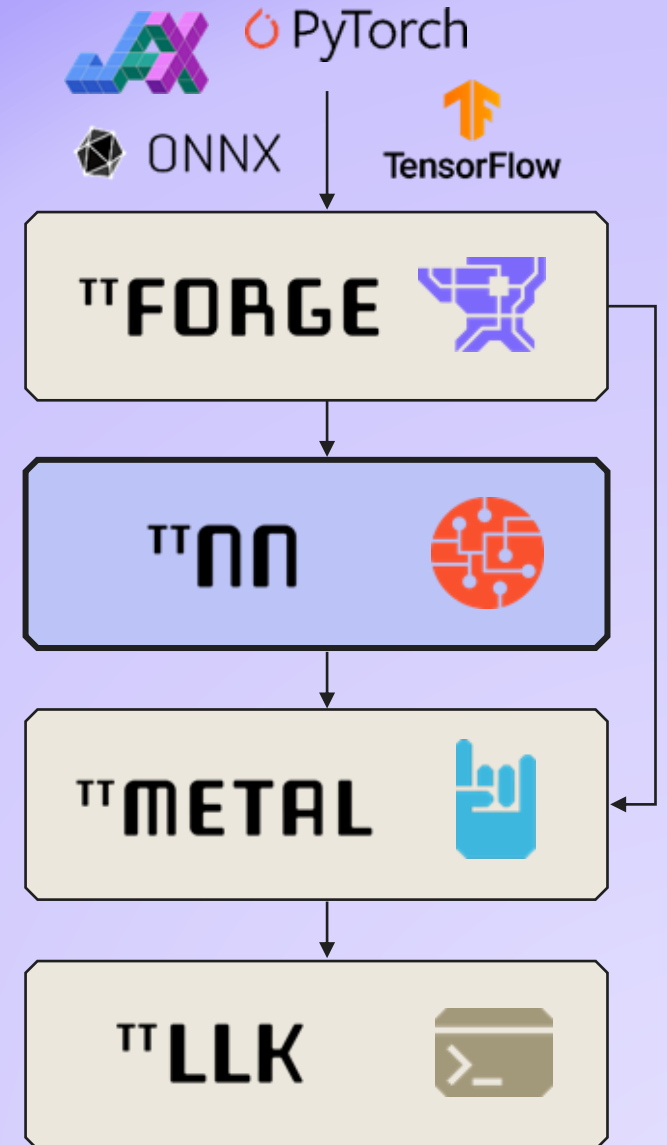
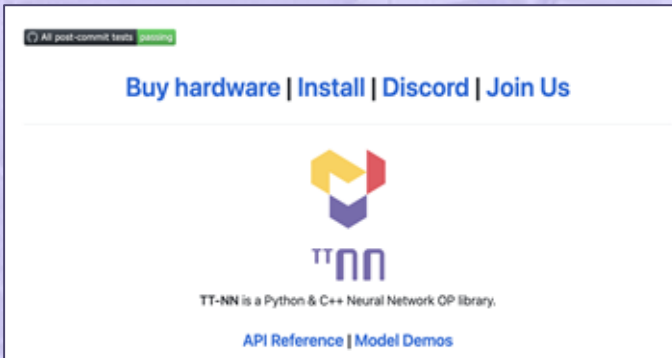
```
import ttnn

# apply rotary
batch, n_heads, _, _ = x.shape

cos = ttnn.repeat(cos, ttnn.Shape([batch, n_heads, 1, 1]))
sin = ttnn.repeat(sin, ttnn.Shape([batch, n_heads, 1, 1]))

x_transformed = ttnn.matmul(x, self.transformation_mat)

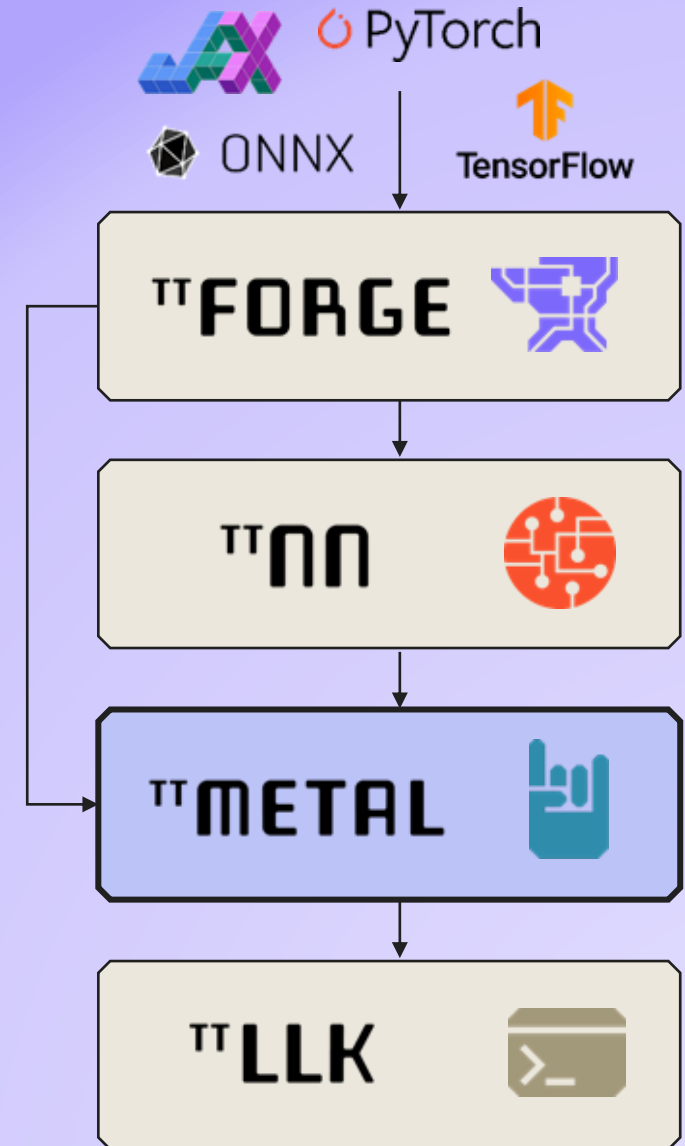
x_cos = ttnn.mul(cos, x)
x_sin = ttnn.mul(sin, x_transformed)
return ttnn.add(x_cos, x_sin)
```



TT-Metalium

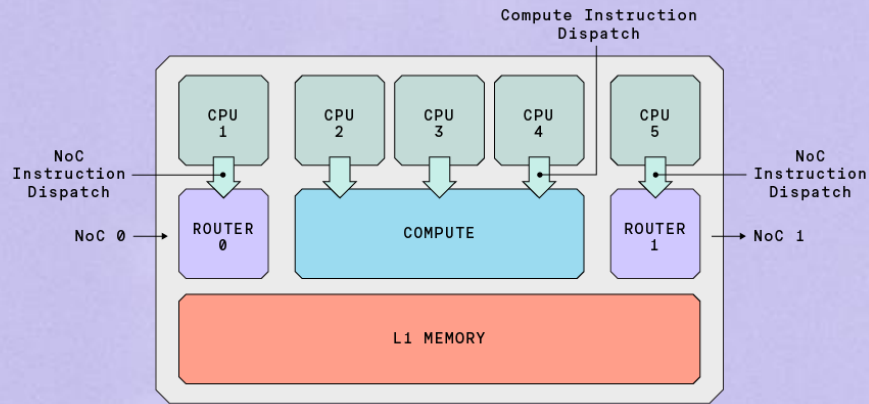
TT-Metalium is a low-level programming model enabling kernel development on Tenstorrent hardware

```
18 namespace NAMESPACE {
19     void MAIN {
20         mm_init(tt::CBIndex::c_0, tt::CBIndex::c_1, tt::CBIndex::c_16);
21         acquire_dst();
22
23         cb_wait_front(tt::CBIndex::c_0, /* number of tiles */ 1);
24         cb_wait_front(tt::CBIndex::c_1, /* number of tiles */ 1);
25
26         matmul_tiles(tt::CBIndex::c_0, tt::CBIndex::c_1, 0, 0, 0, false);
27
28         cb_pop_front(tt::CBIndex::c_1, /* number of tiles */ 1);
29         cb_pop_front(tt::CBIndex::c_0, /* number of tiles */ 1);
30
31         cb_reserve_back(tt::CBIndex::c_16, /* number of tiles */ 1);
32         pack_tile(0, tt::CBIndex::c_16);
33         cb_push_back(tt::CBIndex::c_16, /* number of tiles */ 1);
34
35         release_dst();
36     }
37 } // namespace NAMESPACE
```

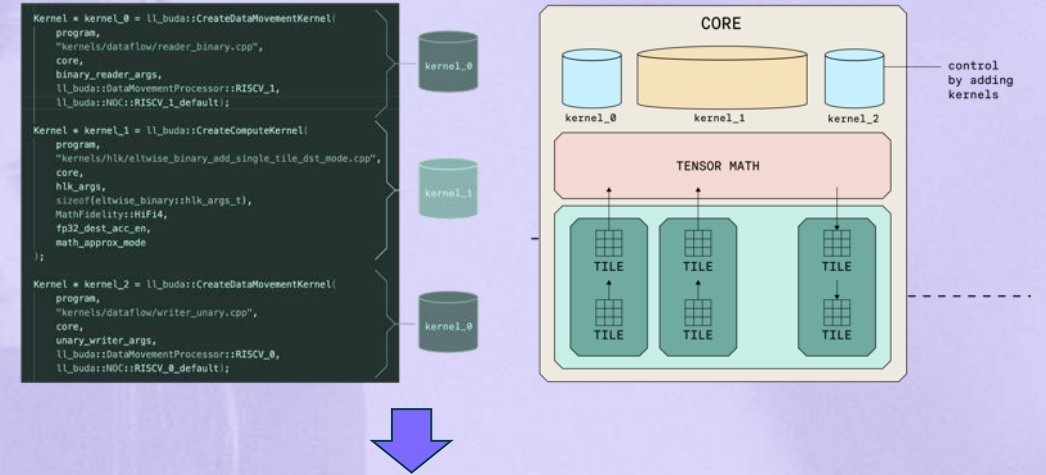


TT-Metalium™: Single Tensix Core to Multi-Chip

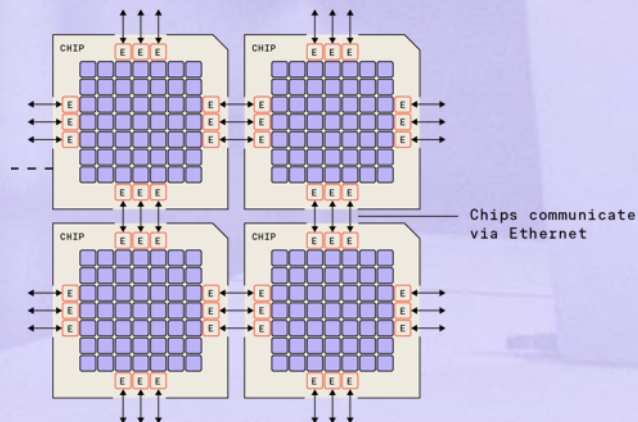
Tensix Core (Direct Access)



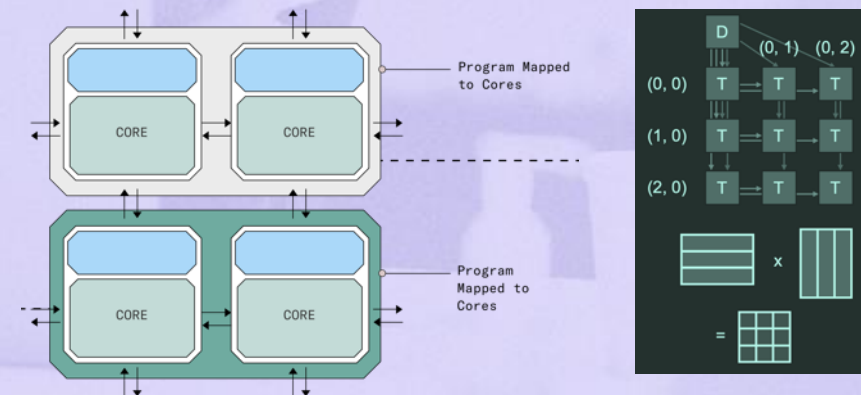
Compute & Data Movement Kernels (Decoupled)



Multi-Chip Scale-Out (Sea of Cores)

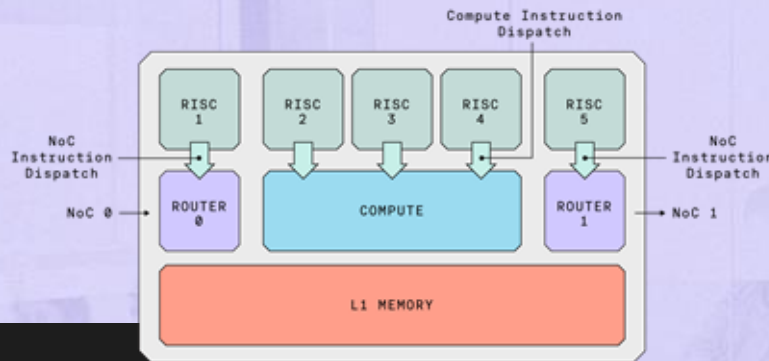


SIMD/MIMD & Multi-Cast (Across Chip)

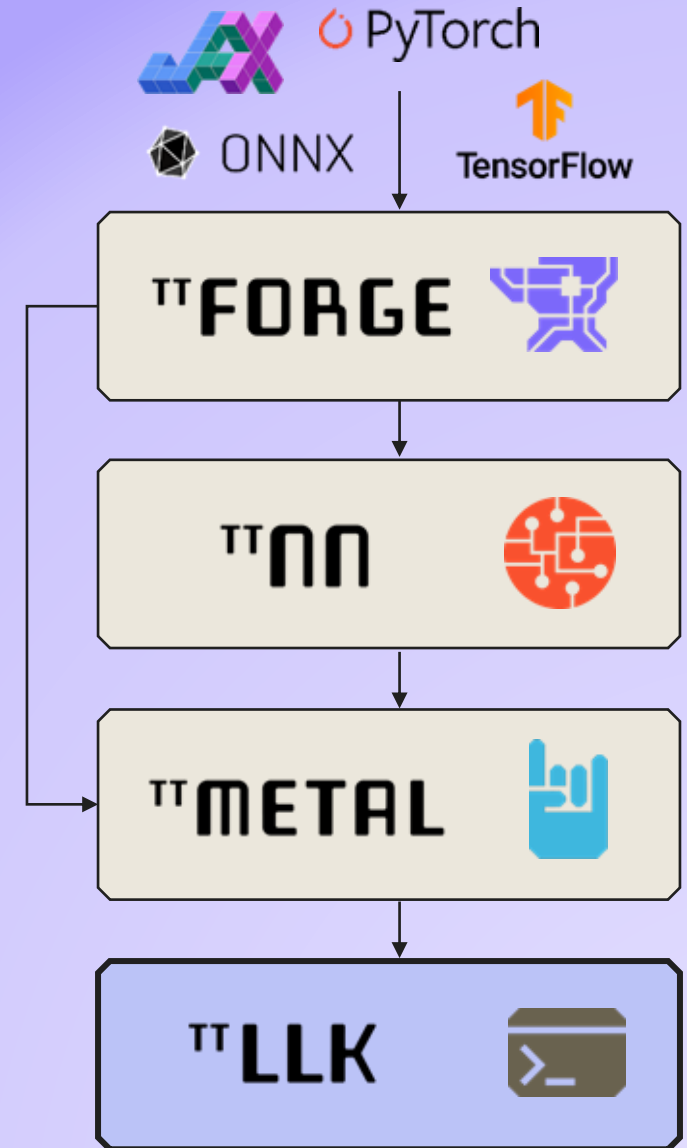


TT-LLK

- Low-level Tensix kernels implementing foundational compute primitives, i.e. matmul, tilize/untile, GeLU, ReLU, add, subtract, TopK etc
- Aims to extract top performance from hardware and deliver it through clean and versatile APIs



```
40 {
41 #ifdef PERF_DUMP
42     if constexpr (MATH_PACK_DECOUPLE)
43     {
44         return;
45     }
46 #endif
47
48 constexpr uint32_t DEST_NUM_TILES_SHIFT = is_fp32_dest_acc_en ? (1) : (0);
49 constexpr uint32_t DEST_NUM_TILES      = DEST_NUM_TILES_FP16 >> DEST_NUM_TILES_SHIFT;
50
51 set_math_semaphores();
52 if constexpr ((Dst == DstSync::SyncHalf) || (Dst == DstSync::SyncTile2))
53 {
54     math_sync_tile_dst_index = 0;
55     dest_section_flip();
56 }
57 else if constexpr (Dst == DstSync::SyncTile16)
58 {
59     math_sync_tile_dst_index++;
60     math_sync_tile_dst_index &= (DEST_NUM_TILES - 1);
61 }
62 }
```



It's All in Github!

Tenstorrent
GitHub



tenstorrent

Type / to search

+

+

+

+

+

+

Overview

Repositories

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90 repositories

Last pushed

tt-metal

Public

TT-NN operator library, and TT-Metalium low level kernel programming model.

ai metal gpu accelerator cuda llama kernels scale-out llm stable-diffusion + 4

C++ Apache License 2.0 259 1.1k 2.7k 574 Updated now

tt-mlir

Public

Tenstorrent MLIR compiler

C++ Apache License 2.0 48 178 775 80 Updated now

tt-llk

Public

C++ Apache License 2.0 18 24 137 22 Updated 1 minute ago

sfpi

Public

C++ Apache License 2.0 12 12 0 1 Updated 1 minute ago

tt-xla

Public



AI Models



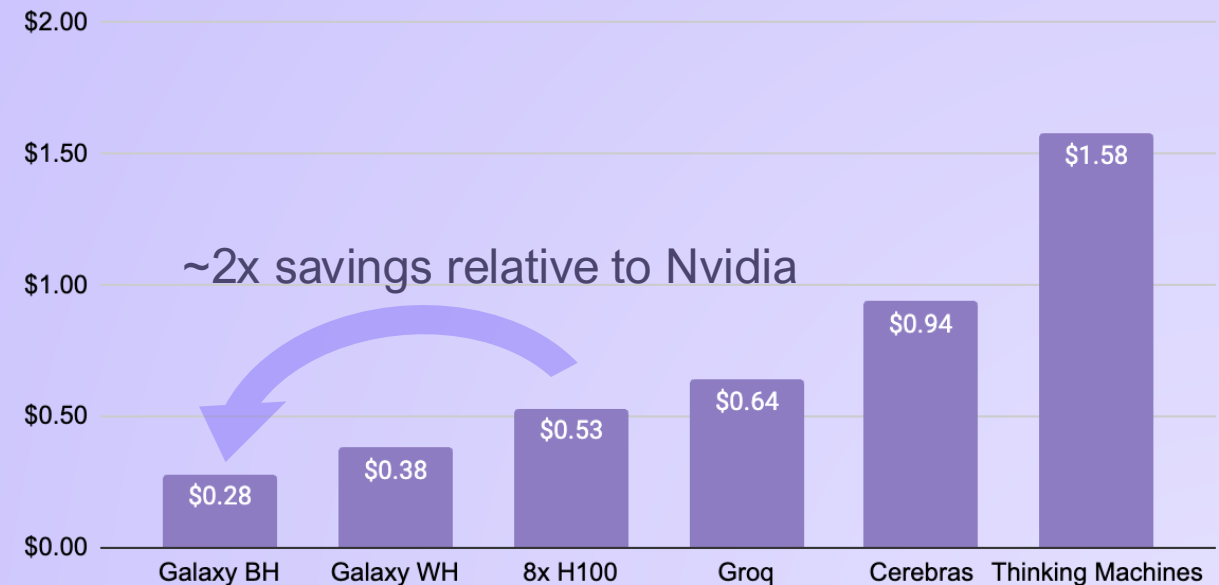
Family	Model				
Dense LLMs	Llama 3.3 70B	✓	✓	✓	✓
	Llama 3.1 8B	✓	✓	✓	✓
	Qwen3-32B	✓	✓	✓	✓
	Qwen/QwQ-32B	✓	✓	✗	✗
	Gemma3-27B	✓	✓	✗	✗
MoE	GPT-OSS-20B	✓	✓	✓	✗
	GPT-OSS-120B	✓	✓	✓	✓
	DeepSeek R1	✓	✓	✗	✗
	Kimi K2	✓	✓	✓	✗
ASR	Whisper	✓	✓	✓	✗
	WhisperX	✓	✓	✓	✗
T2S	PlayAI Dialog	✓	✓	✓	✗
Image	SDXL	✓	✓	✗	✗
	SD3.5	✓	✓	✗	✗
Video	Mochi	✓	✓	✗	✗
	Wan 2.2	✓	✓	✗	✗
Training	Various	✓	✓	✗	✗

Legend: ✓ Available ✗ Not Offered ✗ Limited functionality

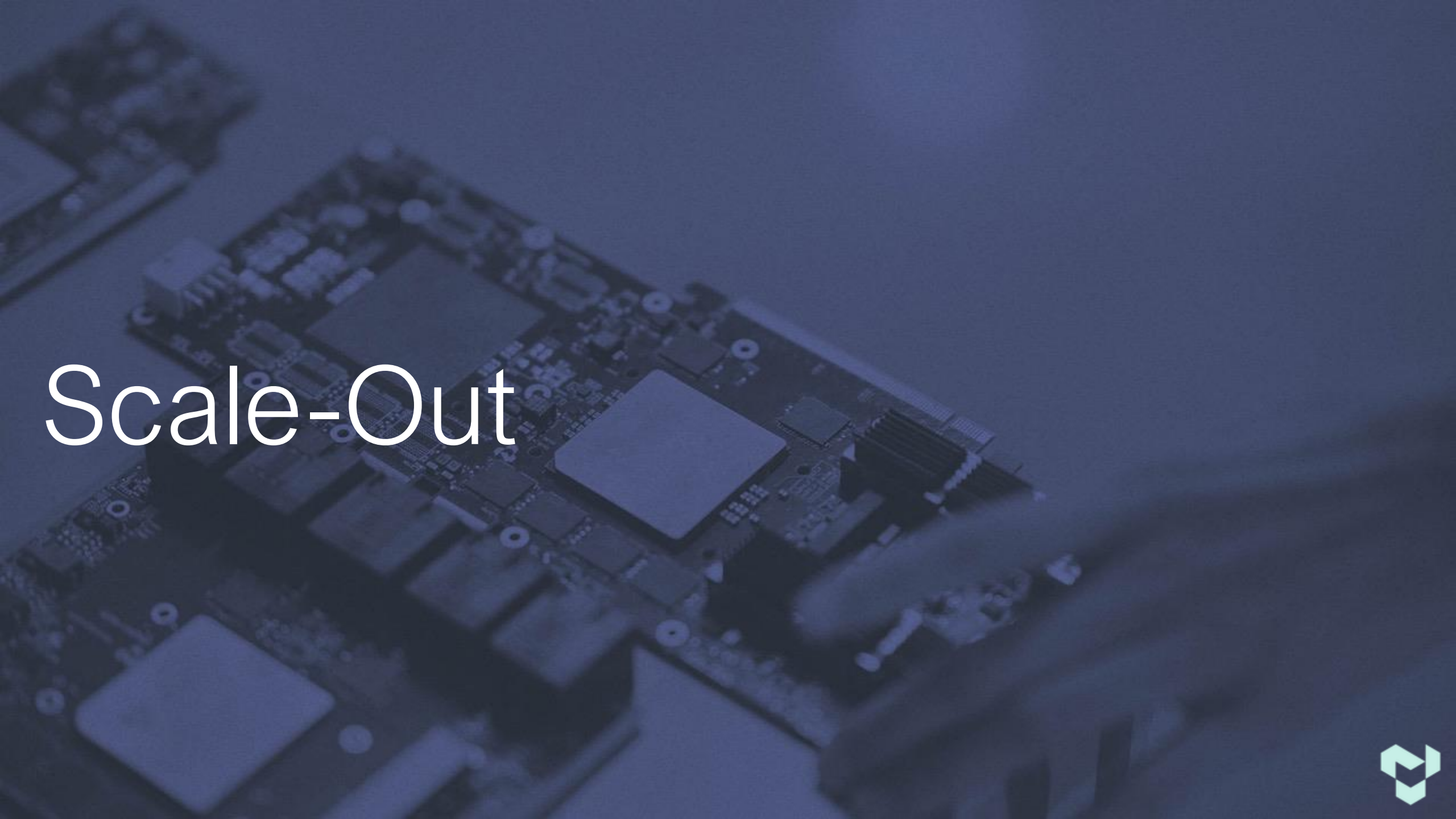
Model Support

- Competitive performance on a broad range of workloads
- LLMs, mixture-of-experts, speech, image & video gen
- Leading on the model frontier with DeepSeek & Wan 2.2
- Lowest unit/token economics

Token Economics (\$/Mtoken)

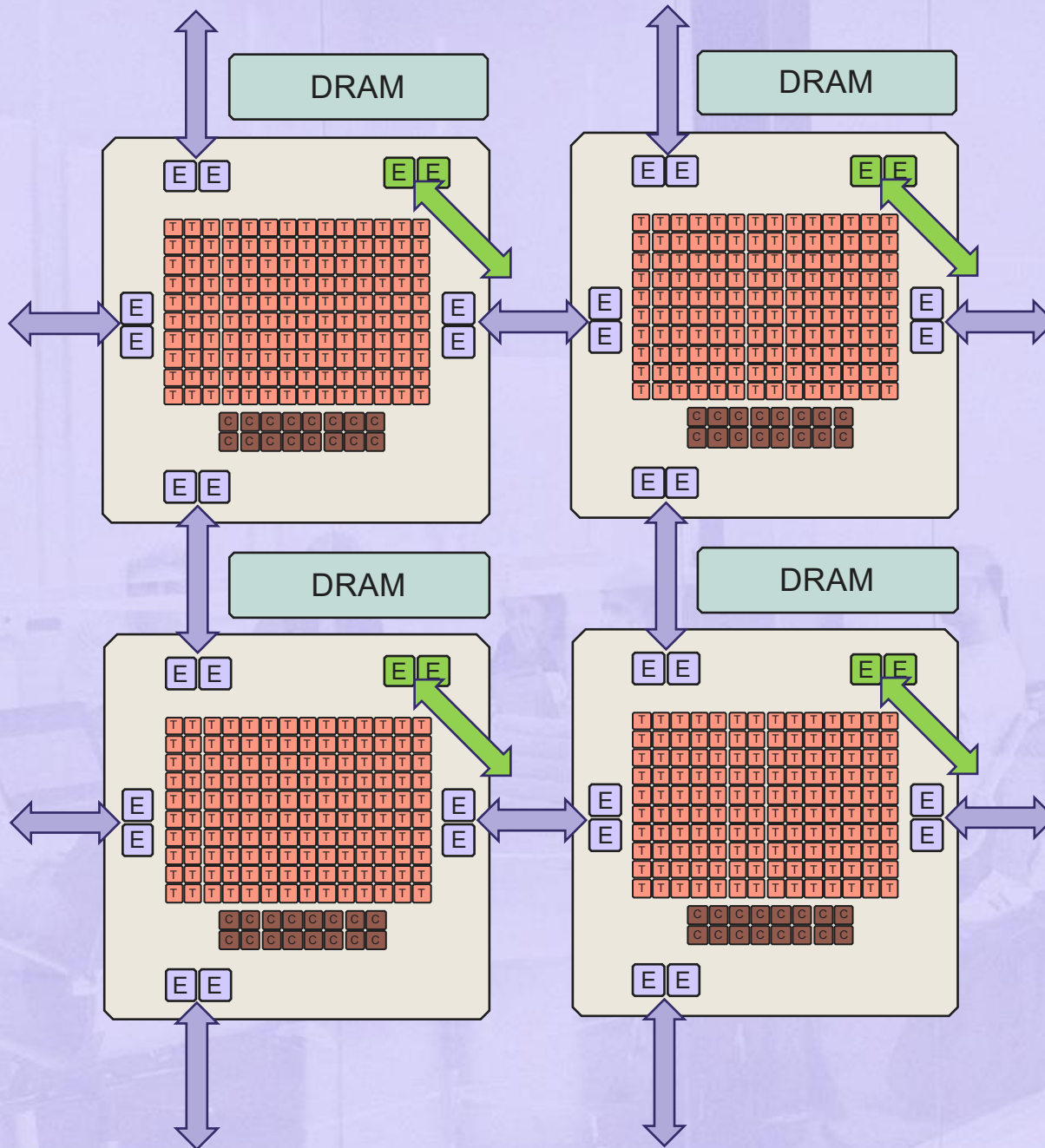


LLama 3.3-70B Batch 32, Seq Len 128, In/Out Mix 3-to-1



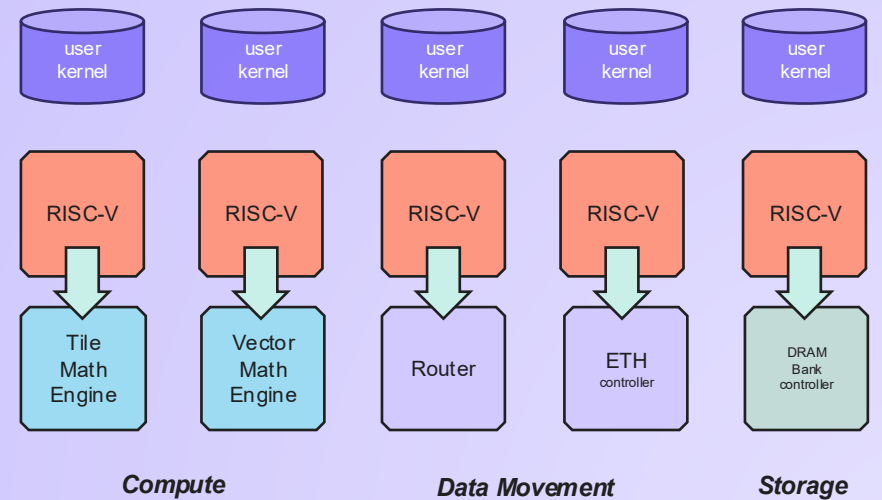
Scale-Out

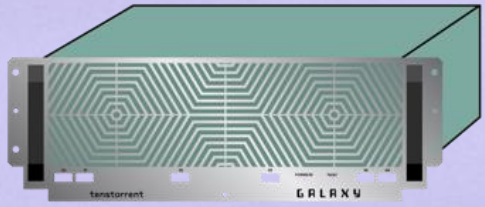




Ethernet Based Scale-Out Blackhole

- 1 TB/s of Ethernet
 - 200 GB/s in N / S / W / E / Z
- Flexible topology
 - 2D / 3D torus
 - Locality and regularity of data movement
 - Sharded data

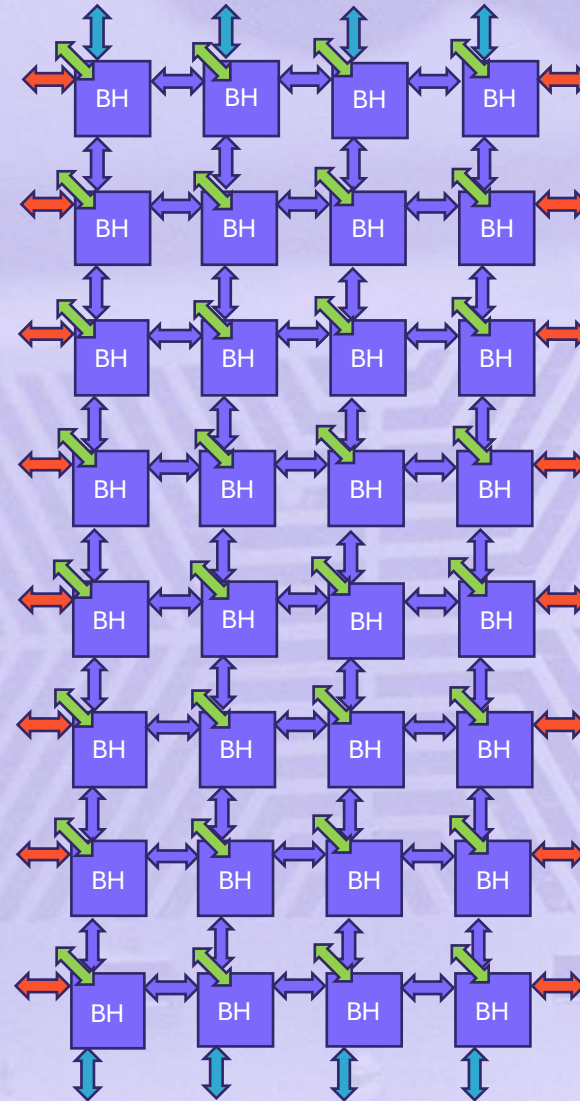




11.2 TB/s Galaxy I/O

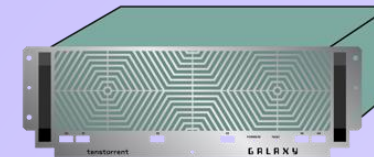
-  X dim I/O: 8 x 200 GB/s
-  Y dim I/O: 16 x 200 GB/s
-  Z dim I/O: 32 x 200 GB/s

I/O out of the box:
 112 x 400 Gbps = total links
56 x 800 Gbps = total ports on the box

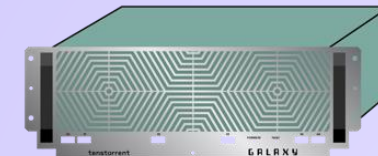


Unified Scale-Out System: Blackhole Galaxy

24 PFLOPs
"AI compute"



11.2 TB/s I/O
"AI switch"



16 TB/s BW, 1 TB capacity
"AI memory"

Thank You

